

CSC 315 Week One TTTT2

Condensed Study Notes

Generated: 2026-05-24 22:54 UTC

CSC 315 : COMPUTER ARCHITECTURE AND ORGANIZATION I

Welcome to Computer Architecture and Organization! This is your first step into understanding how computers work under the hood.

Think of a computer like a very organized office.

In this office, there's a main desk where all the important work happens. This desk is where the computer's 'brain' sits and does all the thinking and processing.

This course will explore the different parts of this 'office' and how they work together to make your computer do what you want it to do.

Week One: Lesson One

Welcome back! We've already touched upon Computer Architecture and Organization as the way we understand how computers are built and how their different parts work together, kind of like understanding how an office building functions.

This week, we're diving into the very core of what makes a computer 'compute'.

Imagine you have a recipe book. This book contains all the instructions for making different dishes. A computer has something similar, called **instructions**. These are like the individual steps in a recipe, telling the computer exactly what to do.

To follow these instructions, the computer needs ingredients. These ingredients are called **data**. Think of data as the raw materials – like flour, sugar, or eggs – that the computer uses to perform its tasks.

So, at its heart, a computer's main job is to take these **instructions** (the steps) and work with **data** (the ingredients) to produce a result. It's a cycle of reading instructions and processing data.

Key ideas:

- **Instructions**: These are the commands that tell the computer what operations to perform.
- **Data**: This is the information or values that the instructions operate on.
- The computer's fundamental process involves executing instructions on data.

Introduction

We've already touched on how computers are built and how their parts work together, like figuring out how an office is organized. We also learned that computers follow instructions to work with data, similar to how a recipe uses ingredients.

Now, let's dive a little deeper into some foundational ideas.

One important concept is the **Von Neumann Architecture**. Imagine a blueprint for a house. This architecture is like a very common and influential blueprint for how most computers are designed internally. It's a way of organizing the computer's main parts so they can work together efficiently. The key idea is that instructions (the steps the computer should take) and data (the information it works on) are stored in the same place, called memory.

When we talk about computers, we're always looking at their **structure** and **function**.

- **Structure** is like the physical parts of a building – the walls, the rooms, the electrical wiring. In a computer, it refers to the actual components and how they are connected.
- **Function** is what those parts *do*. For example, the function of a wall is to separate rooms, and the function of the wiring is to carry electricity. In a computer, function means what each component is designed to accomplish, like processing information or storing it.

So, computer architecture and organization is all about understanding this structure and function, and the Von Neumann Architecture is a major way these pieces are put together in most computers we use.

1.1 Organization and Architecture

You know how we talked about computers following recipes (instructions) to work with ingredients (data)? Well, now we're going to dig a little deeper into *how* that happens by understanding two important ideas: **architecture** and **organization**.

Think of it like building a house. The **architecture** is like the blueprint – it describes what the house will look like from the outside and how the rooms are laid out. It's what you, as someone living in the house, would see and interact with.

- **Architecture** refers to the parts of the computer that a programmer can see and that directly affect how a program runs. This includes things like the set of commands the computer understands (called the **instruction set**), how numbers and letters are represented, and how the computer talks to other devices.
- The **instruction set architecture (ISA)** is a common way to talk about architecture. It defines the format of instructions, the commands themselves (called **opcodes**), the little storage spots for temporary information (called **registers**), and how the computer's memory for instructions and data is set up.

Now, **organization** is like the actual construction and plumbing inside the house. It's about *how* the blueprint is brought to life. You might have two houses with the same blueprint (architecture), but one might have fancy copper pipes, and the other has regular plastic ones. Both work, but they are built differently on the inside.

- **Organization** refers to the specific operational units within the computer and how they are connected to make the architectural design work. It's about the 'behind-the-scenes' details.
- These organizational details are often hidden from the programmer. Think about things like the electronic signals that control different parts, how the computer connects to its printer, or what kind of technology is used for its memory.

Let's use a concrete example:

- **Architecture question:** Should the computer be able to multiply numbers with a single command (a multiply instruction)? This is something a programmer would know about.
- **Organization question:** If we decide to have a multiply instruction, should we build a special, fast circuit just for multiplying? Or should we use the existing 'add' circuit and just have it add numbers repeatedly to achieve multiplication? This is an internal design choice.

Historically, and even today, this distinction is important. Companies often create a 'family' of computers that all share the same architecture (so software written for one will work on others), but they might have different organizations. This means they look and behave the same from a programmer's perspective, but their internal workings and performance might differ based on how they were organized.

In smaller computers like microcomputers, architecture and organization are very closely linked. New technology can lead to new organizational methods, which in turn can allow for more powerful and complex architectures.

1.2 Structure and Function

When we want to understand how something complex, like a computer, is put together and how it works, we can approach it in two main ways: either start with the smallest pieces and build up to the whole system, or start with the big picture and break it down into smaller parts.

It turns out that starting with the big picture and then breaking it down is usually the easiest way to understand complex things. This is because computers, like many other complex systems, are built in layers, or levels.

At each level of a computer's design, we look at two main things:

- **Structure:** This is about how the different parts are connected and related to each other. Think of it like the blueprint of a building, showing how rooms connect and how the foundation supports the walls.
- **Function:** This is about what each individual part actually does and how it operates as part of that larger structure. In our building analogy, function would be what happens in each room – where you sleep, where you cook, etc.

Computers are incredibly complex, with millions of tiny electronic parts. To make sense of it all, we use this idea of hierarchy. A hierarchical system is like a set of nested boxes. Each box contains smaller boxes, and those contain even smaller ones, until you get to the very basic, individual components at the lowest level.

Structure (The Internal structure of the Computer)

Let's explore the internal blueprint of a computer – how its different parts are put together.

Think of a computer like a busy office. It has different departments, and they all need to talk to each other to get work done.

At the heart of this office is the **Central Processing Unit (CPU)**. You can also call this the **processor**. It's like the main manager or the brain of the computer. The CPU is in charge of making sure everything runs

smoothly and also does the heavy lifting when it comes to calculations and data manipulation.

Inside this CPU 'manager's office', there are a few key assistants:

- **Control Unit:** This assistant is like the office manager's personal assistant. It directs the operations of the CPU and, by extension, the whole computer. It tells other parts what to do and when.
- **Arithmetic and Logic Unit (ALU):** This is like the office's dedicated math whiz and logic expert. It's responsible for all the number crunching (like addition, subtraction) and making decisions based on logic (like comparing numbers).
- **Registers:** These are like small, super-fast sticky notes or mini-whiteboards right next to the ALU and Control Unit. They provide very quick storage for data that the CPU is actively working on at that very moment.

Beyond the CPU, there are other important parts in our computer office:

- **Main Memory:** This is like the office's filing cabinet. It's where the computer stores data that it needs to access, but it's not as fast as the registers. It holds more information but takes a little longer to retrieve.
- **I/O (Input/Output):** This represents how the computer interacts with the outside world. Think of it like the mailroom and the reception desk. It's how data gets into the computer (like typing on a keyboard) and how results get out (like displaying on a screen).

Now, how do all these parts communicate? That's where **System Interconnection** comes in. This is like the office's internal phone system or hallway network that allows the CPU, Main Memory, and I/O to send messages and data to each other. A very common way this happens is through a **system bus**, which is like a main highway with many lanes (conducting wires) where data travels between these components.

Similarly, there's **CPU Interconnection**, which is like the internal communication system just within the CPU's 'office', allowing the Control Unit, ALU, and Registers to talk to each other quickly.

Traditionally, computers had just one 'manager' (a single processor). However, modern computers are increasingly using multiple processors working together, like having several managers overseeing different projects simultaneously. Also, the **Control Unit** itself can be designed in different ways. One common method is called **microprogrammed implementation**. This means the control unit works by following a set of very basic instructions, called **microinstructions**, which are like tiny, detailed steps that define exactly how the control unit should behave.

Function

Even though computers can seem really complicated, they actually perform only four basic jobs.

1. **Data Processing:** This is like taking raw ingredients and transforming them into something else. For example, counting items, swapping things around, adding numbers together, or comparing two things to see if they are the same. These are all ways the computer manipulates data.
2. **Control:** Imagine a conductor leading an orchestra. The control unit is like that conductor for the computer. It directs all the different parts of the computer and tells them what to do and when, based on the instructions it receives. It makes sure everything works together smoothly.

3. **Data Storage:** Sometimes, a computer needs to hold onto information. This can be very short-term, like holding the numbers you're currently adding up while you do the calculation. This is called short-term storage. It also needs to store information for longer periods, like saving your documents or photos so you can use them again later. This is long-term storage.

4. **Data Movement:** This involves getting data into or out of the computer. When data comes from a device connected directly to the computer, like a keyboard or a printer, it's called input/output (I/O). The devices themselves are often called peripherals. If data is being sent or received over longer distances, like to another computer on the internet, that's known as data communications.

multicore computer structure

Imagine a computer's main circuit board as a bustling city map. This map, called a **motherboard** (or system board), is a rigid, flat board made of layers that hold and connect all the important electronic parts. Think of it like the city's infrastructure, with copper pathways etched into it, acting like roads for electricity and information.

On this motherboard, you'll find various **chips**. A chip is like a tiny, specially designed building on our city map. It's made from a piece of material like silicon, where many electronic circuits and logic gates are built together. When these circuits are fabricated onto the chip, the whole thing is called an **integrated circuit**.

One of the most important 'buildings' on the motherboard is the **processor chip**. This is where the computer's 'brain' lives. In many modern computers, this single processor chip doesn't just have one 'brain' but several, all working together. When a processor chip has multiple of these 'brains' on it, the computer is called a **multicore computer**.

Each of these individual 'brains' on the processor chip is called a **core**. A core is essentially a processing unit. It's like having multiple chefs in a kitchen, each capable of following a recipe (executing instructions) on their own. A core can be as powerful as a whole CPU on its own, or it might be specialized for certain tasks, like handling complex math calculations.

So, a **processor** itself is the physical chip that contains one or more of these cores. It's the component that reads and carries out instructions. If it has more than one core, it's a **multicore processor**.

These cores and other components on the processor chip, like the **L3 cache** (a small, super-fast memory storage right on the chip for quick access to frequently used data), are connected and managed by control circuits. While a diagram might show the L3 cache in separate sections, usually all the cores can access the entire L3 cache. The layout of these processor chips can vary, but they generally aim to fit many cores and cache memory onto a single piece of silicon.

Besides the processor chip, the motherboard also has slots for other important components, such as memory chips (where the computer stores information long-term) and I/O controller chips (which manage how the computer talks to the outside world, like your keyboard and screen). These are like different districts or specialized buildings in our city, each with its own function.

Inside a Single Processor Core

Imagine a processor chip as a bustling city, and each 'core' is like a specialized neighborhood within that city, responsible for a specific set of tasks. This section focuses on what's inside one of these neighborhoods.

At the heart of a core, you'll find the 'Arithmetic and Logic Unit' (ALU). Think of the ALU as the core's calculator and decision-maker. It's the part that actually performs mathematical calculations (like adding numbers) and logical operations (like comparing if one thing is bigger than another), based on the instructions it receives.

Another crucial part is the 'Instruction Logic'. This is like the core's librarian. It's responsible for fetching instructions from memory, like pulling a book off a shelf. Then, it 'decodes' each instruction, which means figuring out what the instruction is asking it to do and where to find the necessary 'operands' (the data or values it needs to work with).

To speed things up, each core has its own 'L1 cache'. Think of the cache as a small, super-fast desk right next to the librarian and the calculator. It holds frequently used instructions ('I-cache') and data ('L1 data cache') so the core doesn't have to go all the way to the main memory (like going to a large, slower library) every single time. Many cores also have an 'L2 cache', which is like a slightly larger, but still very fast, filing cabinet nearby.

There's also 'Load/store logic'. This is like the core's messenger service. It handles the movement of data back and forth between the core's fast cache and the main memory. It makes sure data gets to where it needs to be quickly and efficiently.

While the core handles the processing, the rest of the computer system has other parts to manage connections. For example, 'Ethernet controllers' and 'Ethernet ports' are like the core's internet and network connection points, allowing it to communicate with other computers. 'PCI-Express slots' are like high-speed highways for connecting powerful devices like graphics cards, and 'Serial ATA (SATA) sockets' are used to connect storage devices like hard drives. 'USB sockets' are the common connection points for many everyday devices.

1.3 Von Neumann architecture

Imagine a very organized chef who follows a recipe book step-by-step, using the same kitchen counter for both the recipe instructions and the ingredients. This is similar to how a Von Neumann computer works.

In the 1940s, a mathematician named John Von Neumann described a fundamental way to arrange a computer's parts. This arrangement is called the Von Neumann architecture.

A Von Neumann computer works by:

- Carrying out instructions one after another, in a specific order. This is like the chef doing one step of the recipe, then the next, and so on.
- Using a single memory (and a single pathway to access it) for both the program instructions and the data it needs to work with. The computer doesn't inherently know if something in its memory is an instruction or data; it figures this out based on where it is in the memory.
- Typically using just one main processing unit (or processor) to do all the work.

Most computers we use today are based on this Von Neumann concept, even though there are other types of computer designs out there. When we talk about the Von Neumann architecture, we're really discussing how the different hardware pieces are connected and work together in this specific type of computer.

Von Neumann bottleneck

Imagine a very busy highway where all the cars (instructions and data) have to travel down the same single lane. No matter how fast the cars are, they can only go one after another. This is similar to how a computer, following the Von Neumann design, can only do one thing at a time.

Even if you make the processor (the part of the computer that does the thinking and calculating) super-fast, or give it a huge amount of RAM (main memory) or cache (super-fast temporary storage), it can still get held up.

This limitation, where the speed of processing is limited by how quickly instructions and data can be moved to and from the processor, is called the 'Von Neumann bottleneck'. It's like that single-lane highway holding back all the faster cars.

Key takeaway:

- The Von Neumann architecture, where instructions and data share the same path, can create a slowdown, even with fast components.